



TEMPLATE-DESIGNED ORGANIC ELECTRONICS
RESEARCH TRAINING GROUP 2591

M. Sc. Kristin Gratzfeld
Institute of Physical and Theoretical Chemistry
University of Bonn
Wegelerstraße 12
53115 Bonn

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C1: Self-organized structures of organic molecules on electrode surfaces

TIDE project ID: C1

Supervisor: Prof. Dr. Moritz Sokolowski

Co-Supervisor: Prof. Dr. Sigurd Höger

1 State of the art & preliminary work

1.1 State of the art

Ultrathin organic films are important prerequisites for many recent technological applications. They play role for sensors, corrosion inhibition, nanoelectronics, batteries, and nano-composites [1, 2]. Especially, organic films of peri-arylene molecules have been studied widely due to their high electrical conductivity, chemical stability, and strong fluorescence yields [3, 4]. Symmetric peri-arylene molecules form stable intermolecular frameworks with well-defined rigid structures on surfaces [5].

One of the most studied peri-arylenes is **PTCDA** (perylene tetracarboxylic dianhydride). In the past, many growth studies of vacuum-deposited thin self-organized films of **PTCDA** on metal surfaces were performed under ultrahigh vacuum (UHV) conditions [6, 7]. However, there are many fragile peri-arylene systems, such as perylene tetracarboxylic acid (**PTCA**) that cannot be thermally evaporated and are thus not suitable for vacuum deposition on surfaces. One possibility to achieve ordered thin organic films of these fragile peri-arylenes on surfaces is the deposition from solutions [8] or the electrochemical deposition [9]. The peculiarity of electrochemical deposition of molecules on surfaces is that the adsorption is controlled by the electrode potential. The orientation and order of the electrochemically adsorbed molecules depend on the applied electrochemical potential. This aspect can be used to optimize the structural and electronic properties of the organic film structures at the solid/liquid interface.

In the present project, we consider the electrochemical adsorption of PTCA and other organic π conjugated molecules on metal surfaces, in particular on the Au(111) surface. The aim of the project is also to investigate in how far the structures of **PTCA** on Au(111) compare with those well-known structures of the **PTCDA** on Au(111) [10]. The vision of the project is that electrochemically grown films can be used as templates for the further deposition of semiconducting organic molecules. In this context, it is worth to note that **PTCA** is presently used as a surfactant for the growth of the 2D-material MoS₂ [11, 12] and positively charged gold nanoparticles (NP) [13].

1.2 Preliminary work

Figure 1 illustrates the synthesis of **K₄PTCA**. For this purpose, **PTCDA** was dissolved in aqueous KOH solution. By this process the anhydrides are opened and the fully deprotonated **PTCA⁴⁻** anion is formed. The **PTCA⁴⁻** anion is completely solvable in water [14] because of the repulsive electrostatic forces between the anions. A yellow fluorescent solution is obtained [15]. In acidic conditions (e.g., by addition of CH₃COOH or HClO₄) the **PTCA⁴⁻** anion is completely protonated and forms aggregates of **H₄PTCA_(s)** that are not soluble in the aqueous solution [14].

After the **PTCA⁴⁻** had formed, ethanol was added to the solution [15]. This causes that the salt **K₄PTCA** precipitates as a yellow powder. It was subject to centrifugation, washing by ethanol, and

drying at 80°C under vacuum. For the further experiments **K₄PTCA** was dissolved in 0.1 M aqueous KClO₄ solution yielding a concentration of 1 mM. The resulting pH is about 7-8.

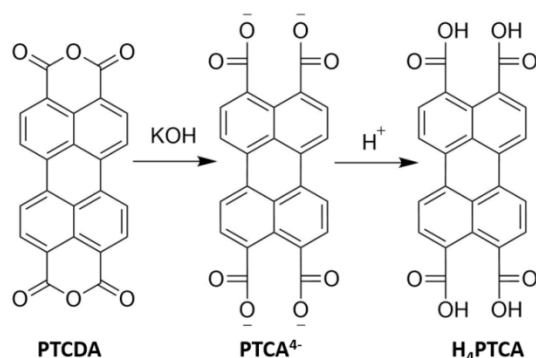


Figure 1. Chemical reaction of the PTCA⁴⁻ formation and protonation (H₄PTCA).

We further focused on the adsorption of **PTCA⁴⁻** layers on surfaces. To this end, we investigated the adsorption of **PTCA⁴⁻** on a potential controlled Au(111) electrode surface by cyclic voltammetry (CV) and electrochemical scanning tunneling microscopy (EC-STM).

CV is a suitable tool for fast characterization of the electrode/electrolyte interface. In the experiment, the change in the current of the electrode is recorded while the electrode potential is scanned linearly in time. The CV measurements shown in Figure 2 were performed with a freshly prepared Au(111) crystal in a pure 0.1 M KClO₄ solution (black curve). When 1 mM **K₄PTCA** was added to the 0.1 M KClO₄ solution, adsorption of **PTCA⁴⁻** occurs at 210 mV in the anodic direction (− → +) and desorption at 64 mV in the cathodic direction (+ → −) (red curve). It is also noticeable that the intensity of the adsorption peak is much higher than that of the desorption peak. The reason for this is not clear, yet.

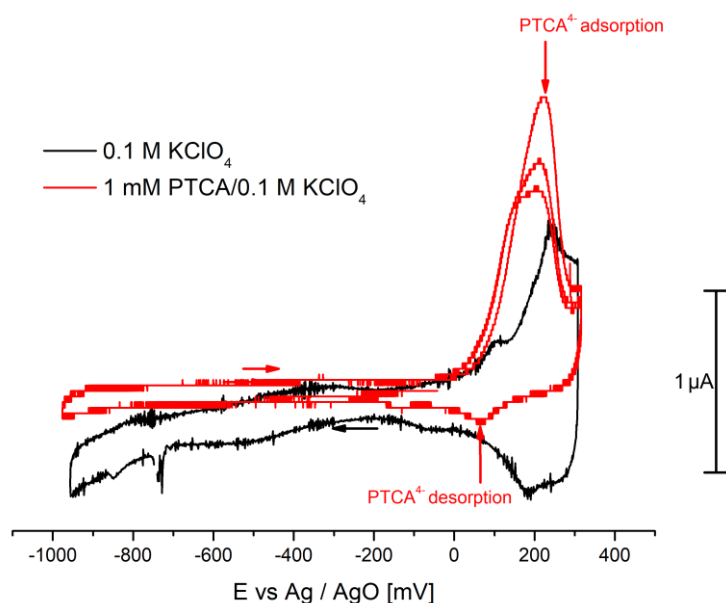


Figure 2. Comparison of CV curves recorded for 0.1 M KClO₄ (black) and 0.1 M KClO₄ + 1 mM **PTCA** (red). Sweep rate 10 mV/s.

To obtain more information about the morphology of **PTCA**⁴⁻ anions on the potential controlled Au(111) surface, EC-STM measurements were performed. **Figure 3** shows a series of EC-STM images of 1 mM **PTCA**⁴⁻ in aqueous 0.1 M KClO₄ solution at different surface potentials. At positive potentials, cluster formation is observed on the surface. For the negative potentials, the clusters disappear again.

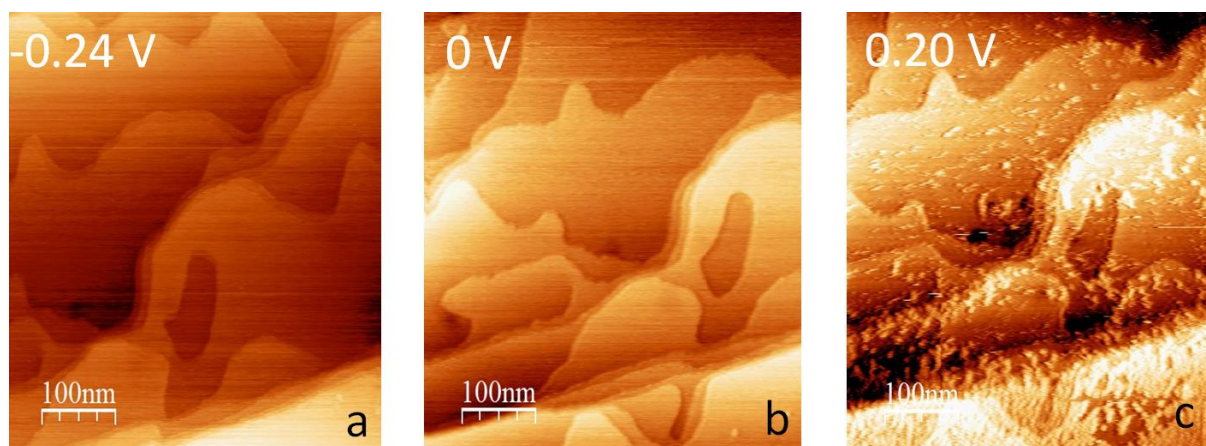


Figure 3. a-c: EC-STM images of the Au(111) surface in 10^{-3} M of K₄PTCA in 0.1 M KClO₄ for different electrochemical potentials (E). Size of the image $(500 \text{ nm})^2$, $U_{\text{bias}} = 0.15 \text{ V}$, $I_t = 0.2 \text{ nA}$, **a** $E = -0.24 \text{ V}$, **b** $E = 0 \text{ V}$, **c** $E = 0.20 \text{ V}$. (U_{bias} refers to the tip.)

2 Research vision & project idea

There are several points that need to be considered in order to finally obtain self-ordered peri-arylene structures on potential controlled Au(111) surface. The first step is to understand the morphology of the adsorbed **PTCA**⁴⁻ molecules on the surface by performing potential adsorption.

As described above, the **PTCA**⁴⁻ forms disordered clusters on the positively charged Au(111) surface. We explain this by adsorption of the COO⁻ groups at the Au(111) surface [16]. This is illustrated in Figure 4. Due to the electrostatic repulsion of the terminal COO⁻ groups of **PTCA**⁴⁻ oriented in the direction of the electrolyte it is possibly difficult that the anions form self-ordered structures on the surface. Hence, it is a general question, whether it is possible to achieve ordered structures of **PTCA**⁴⁻ on the potential-controlled surface at all.

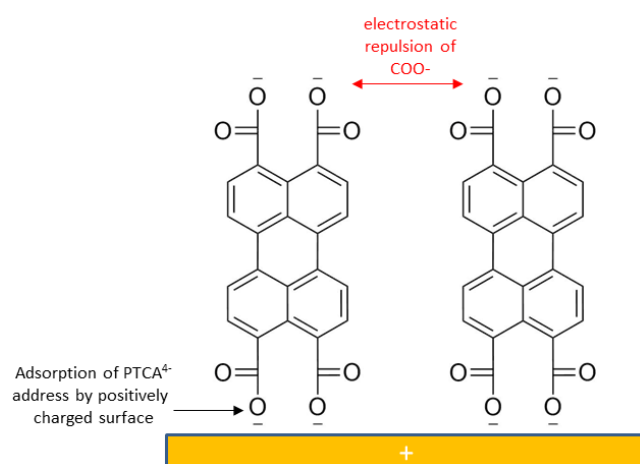
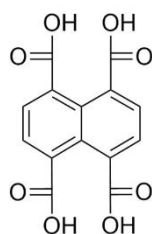


Figure 4. Model for the disordered layers of **PTCA**⁴⁻ on Au (111) surface.

In order to clarify this, an experiment of a known system, with a molecule without terminal COO⁻ groups will be performed. The adsorption of benzoic acid on potential controlled surface was already investigated several times by EC-STM [9, 16]. This experiment will be repeated and will serve as a control experiment.

The charge of the terminated COO⁻ groups of **PTCA**⁴⁻ is related to the deprotonation/protonation and depends on the pH value of the electrolyte [14]. Decreasing the pH-value of the electrolyte leads to protonation of the acidic groups of **PTCA**⁴⁻ and formation of **H₄PTCA**. Upon protonation of the acid groups, also hydrogen-hydrogen bond interactions may occur between the **H₄PTCA** molecules. These would possibly stabilize the laterally ordered structures of **H₄PTCA** on the Au(111) surface. In addition, this could lead to planar adsorption of **H₄PTCA** on the Au(111) surface. This orientation of **H₄PTCA** promotes hydrogen interactions with neighboring **H₄PTCA** molecules and formation of a molecular network on the Au(111) surface. In this case, **H₄PTCA** would be completely fixed due to the lateral interactions on the Au(111) surface. In order to obtain an overview of the adsorption behavior during protonation, we plan to perform pH-dependent CV measurements.

Since studies on pH-dependent protonation are already available for naphthalene-teracarboxylic acid (**H₄NTCA**, see Figure 5) [17], it would be useful to perform pH-dependent CV and EC STM measurements for this molecule in addition. Due to the smaller π -conjugated core of **H₄NTCA** the adsorption on Au(111) could, however, differ from **H₄PTCA**.



H₄NTCA

Figure 5. Chemical structures of **H₄NTCA**.

As mentioned above, the formation of the adsorbed **H₄PTCA** structures on Au(111) can occur by electrochemical deposition and deposition from a solution without potential control. For the formation of solution deposited self-ordered **H₄PTCA** structures, the Au(111) crystal will be immersed in aqueous **PTCA**⁴⁻ solution for a few minutes and washed with HClO₄ solution. Here, it would be interesting to perform STM measurements on these self-assembled **H₄PTCA** structures deposited from the solution on the Au(111) surface in addition and to compare them to structures of **H₄PTCA** obtained from electrodeposition. Due to the pH-dependent optical properties [15] of **H₄NTCA/H₄PTCA** optical spectroscopy can be performed in addition.

Based on the experience above, additional EC-STM measurements could be performed with other organic π conjugated molecules on metal surfaces. Due to the zwitterionic properties of squaraines, that are also investigated in TIDE project A1 these are attractive molecules to be considered for further investigations.

3 Project milestone & working plan for the next 6 months

3. 1 General timeline of the project

The project is going on for about one year. The results of the first year are partly described in the preliminary work. The work of the first 6 months are described below under **A** and **B** and that of the following months under **C**.

3. 2 Working plan for the next six month and milestones

The project is divided into three sub-projects. **Block A** contains the preliminary experiments. The preliminary experiments are important basic experiments for testing the equipment and to ensure reproducible experiments. The preliminary experiments include the preparation of the Au(111) crystal for achieving a clean and flat electrode, and practicing clean working under argon inert gas atmosphere for reversible electrochemistry in the STM cell. This will be **Milestone 1**. In addition, the preparation of the isolated STM tips with good resolution will be trained. These basic experiments are performed using a potential controlled Au(111) crystal in 0.1 M H₂SO₄ (**A1**). A second preliminary experiment is the reproduction of a known system from the literature (**Milestone 2**). Here the adsorption of benzoic acid (**A2**) on potential controlled Au(111) surface will be investigated [16]. COO⁻ groups of benzoic acid (BA) are coordinated to the positively charged Au (111) surface and form self-organized sub molecular structures [16].

In **block B** the adsorption of NTCA on potential controlled surface Au(111) surface is investigated. In **B1** pH-dependent CV measurements will be performed to have an overview of the adsorption NTCA⁴⁻/H₄NTCA on the potential controlled Au(111) surface (**Milestone 3**). In **B2** EC STM measurement of NTCA and H₄NTCA/Au(111) will be performed (**Milestone 4**).

Table 1. Timetable of the working process for the next 6 months

Months	Jul. 21	Aug. 21	Sep. 21	Okt. 21	Nov. 21	Dec. 21
	A1	A2	A2			
			B1	B2		
					C1/C2	C2

3. 3 Working plan for further time and milestones

In **block C**, all experiments with PTCA as the surface adsorbate are included. Here we focus on the differences between *electrodeposition* and *solution deposition* of organic films on the Au(111) surface.

In **C1** self-assembled H₄PTCA self-organized monolayers deposited from solution are investigated on graphite. For this project, the anion is adsorbed from alkaline solution. In a second step, the anion is protonated on the surface by exposure to HClO₄.

As a second step, **H₄PTCA** pH-dependent CV measurements on potential controlled Au(111) will be performed (**C2**) (**Milestone 5**). In **C3** EC-STM investigations on **PTCA⁴⁻/H₄PTCA** on Au(111) will follow (**Milestone 6**).

Due to the photoluminescent properties of **PTCA/NTCA** the pH-dependent measurements will be also followed by optical spectroscopy in parallel.

In **C4** CV measurements and EC-STM investigations of squaraines on Au(111) surface will be performed. Due to the optical properties of the squaraines optical spectroscopy measurements will be conducted in addition (**Milestone 7**).

Table 2. Timetable of the working process for the final 14 month.

Quartal/ year	I/2022	II/2022	III/2022	IV/2022	I/2023	II/2023	III/2023	IV/2023	I/2024
	C3	C3	C3	C4	C4	C4	Evaluation	Writing	Writing

3. 4 Research resentence abroad

Due to the COVID crisis, the plan for the research stay abroad is not finally fixed in time, yet. We plan to perform the project C3 within the research resentence. For the C3 project, EC STM measurements will be implemented in the group of Prof. Dr. Marek Nowicki at the Institute of Experimental Physics at the University of Wroclaw/Poland. A collaboration with this group within a DAAD project exists.

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